Forecasting Indian Stock Returns from Supplier Stocks with Machine Learning

UE24CS352A — Machine Learning Mini-Project | Team: Nyssa, Yathindra

1. Problem statement

Can the past returns of Maruti Suzuki's suppliers, together with Maruti's own price and volume history, predict its return over the next five trading days better than simply guessing “no change”? And does a trading rule built on those forecasts beat buy-and-hold once transaction costs are included? We adapt R. M. Glawion's Stanford CS229 report *Quantitative Trading with Machine Learning* (2021), which predicted Volkswagen's returns from 30 suppliers' lagged returns, found Elastic Net best, and reported a 21.49% strategy return vs 7.98% for buy-and-hold. We apply the idea to an Indian automaker and address four limitations of the report: no baseline, missing prices filled with zero, no transaction costs, and no risk measures.

2. Dataset

Daily split- and dividend-adjusted prices from Yahoo Finance via `yfinance` (10 years to 5 Oct 2026). **Target:** `MARUTI.NS`. **Peers:** auto components (`MOTHERSON, BOSCHLTD, UNOMINDA, ENDURANCE`), tyres/batteries (`MRF, APOLLOTYRE, EXIDEIND`), steel (`TATASTEEL, JSWSTEEL`) and the NIFTY 50 (`^NSEI`). 2,471 trading days give **2,415 usable rows (23 Dec 2016 – 5 Oct 2026)**: the first 50 are dropped for moving-average warm-up and the last 5 because they have no future price. All peers covered at least 90% of days; holiday gaps were forward-filled for at most 5 days using past prices only. Nothing was filled with zero.

3. Approach

- **Features at day t** (data up to t only, 61 in total): 11 own features (1/5/20-day returns; price vs 5/20/50-day moving averages; 10/20-day volatility; volume change; intraday range; close location within the day's range) plus each peer's daily return on each of the last 5 days.
- **Target:** `Close[t+5]/Close[t] − 1`; horizons of 1 and 20 days also tested.
- **Models:** Elastic Net, Decision Tree, Random Forest, Gradient Boosting; **baselines:** predict 0 and historical mean.
- **Split:** chronological. Train Dec 2016–Oct 2022 (1,444 rows), validation Nov 2022–Oct 2024 (478), test Oct 2024–Oct 2026 (483). The 5 rows before each boundary are dropped (embargo) so no training target overlaps a later period.
- **Selection:** each model is tuned on validation RMSE and the best is chosen on validation; all are refit on train+validation and scored once on test.
- **Metrics:** RMSE, MAE, direction accuracy, and skill vs predict-0 = $1 - \mathrm{SSE}_{\mathrm{model}}/\mathrm{SSE}_{\mathrm{zero}}$.
- **Robustness:** ablation without peer features; 5-period walk-forward retraining on all earlier data; 1/5/20-day horizon comparison.
- **Backtest:** long/flat. Hold Maruti on day t if the forecast made at close t–1 exceeds 0.25%, else hold cash; each buy or sell costs 0.05%.

4. Implementation

`trading_ml.py`: cleaning, peer alignment, features, splitting, tuning, metrics, ablation, walk-forward and backtest. `app.py`: Streamlit dashboard for the live demo (Yahoo Finance, CSV upload or simulated data). `run_experiment.py`: reproduces every table and chart into `reports/results/`. `tests/test_pipeline.py`: 11 tests covering look-ahead leakage, the target, the embargo, backtest timing and the ablation.

5. Results

Model	RMSE own only (%)	RMSE own + peers (%)	MAE (%)	Direction acc. (%)	Skill vs predict-0 (%)
Baseline: predict 0	3.375	3.375	2.493	—	0.0
Random Forest	3.432	**3.356**	2.511	53.6	**+1.1**
Gradient Boosting	3.420	3.385	2.517	52.0	–0.6
Elastic Net	3.387	3.387	2.518	48.7	–0.7
Baseline: historical mean	3.387	3.387	2.518	48.7	–0.7
Decision Tree (selected)	3.470	3.470	2.578	51.6	–5.7

Table 1. Test set (Oct 2024 – Oct 2026), 5-day horizon; lower RMSE is better. MAE, direction and skill columns are for own + peers.

- **Accuracy:** only Random Forest with peers beat predict-0 on test, narrowly. Validation RMSEs were nearly tied (2.702–2.762%), and the model picked on validation (Decision Tree) was the worst on test. In walk-forward, the historical-mean baseline had the lowest average RMSE (3.335%); Random Forest and Gradient Boosting beat predict-0 in 2 of 5 periods, Elastic Net in 1, Decision Tree in 0.
- **Peers:** lowered RMSE for Random Forest (–0.076 pts) and Gradient Boosting (–0.036); Elastic Net and Decision Tree effectively ignored them. The Decision Tree's only peer split was on Motherson's previous-day return, in a high-volatility branch the test period never reached.
- **Horizons:** predict-0 RMSE grows from 1.44% (1 day) to 3.38% (5) and 7.54% (20). At 20 days, Gradient Boosting (7.31%) and Random Forest (7.36%) with peers beat predict-0; without peers only Gradient Boosting did, marginally (7.51%).
- **Strategy** (Decision Tree): **+6.7% vs** –**3.3%** for buy-and-hold; max drawdown –23.1% vs –33.5%; Sharpe 0.28 vs 0.04; 32 trades; 71% of days invested. The gap comes mainly from holding cash during the late-2026 decline (Figure 1). With own features only, the return ranged from –3.2% to +24.7% depending on the threshold.

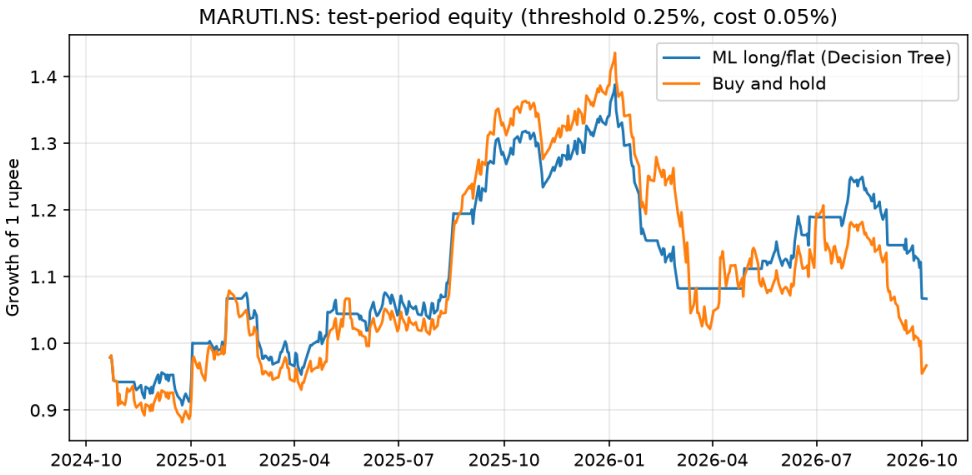


Figure 1. Test-period growth of 1 rupee: ML long/flat (Decision Tree, threshold 0.25%, cost 0.05%) vs buy-and-hold.

6. Conclusion

Supplier returns carry a small amount of information about Maruti's future returns, and only the non-linear ensembles could use it. It is not enough to reliably beat guessing “no change”: one model did so on test, none did on average in walk-forward testing, and the gains were a fraction of a percentage point. This agrees with the reference paper's own observation that predicting zero gives almost the same RMSE. The strategy beat buy-and-hold, but the forecasts behind it show no reliable edge and the result swings with the threshold, so we treat it as an illustration of how backtests can mislead rather than evidence of a profitable signal. The same caution applies to the paper's headline result, which was the best of 96 configurations and ignored costs.

Limitations: hand-picked supplier list (not verified contracts); one validation period with small tuning grids; no slippage or taxes; a single target stock; correlation, not causation. **Future work:** verified supply-chain data, macro variables (bond yields, INR/USD), re-tuning within each walk-forward period, and other Indian automakers.